

Applied NLP

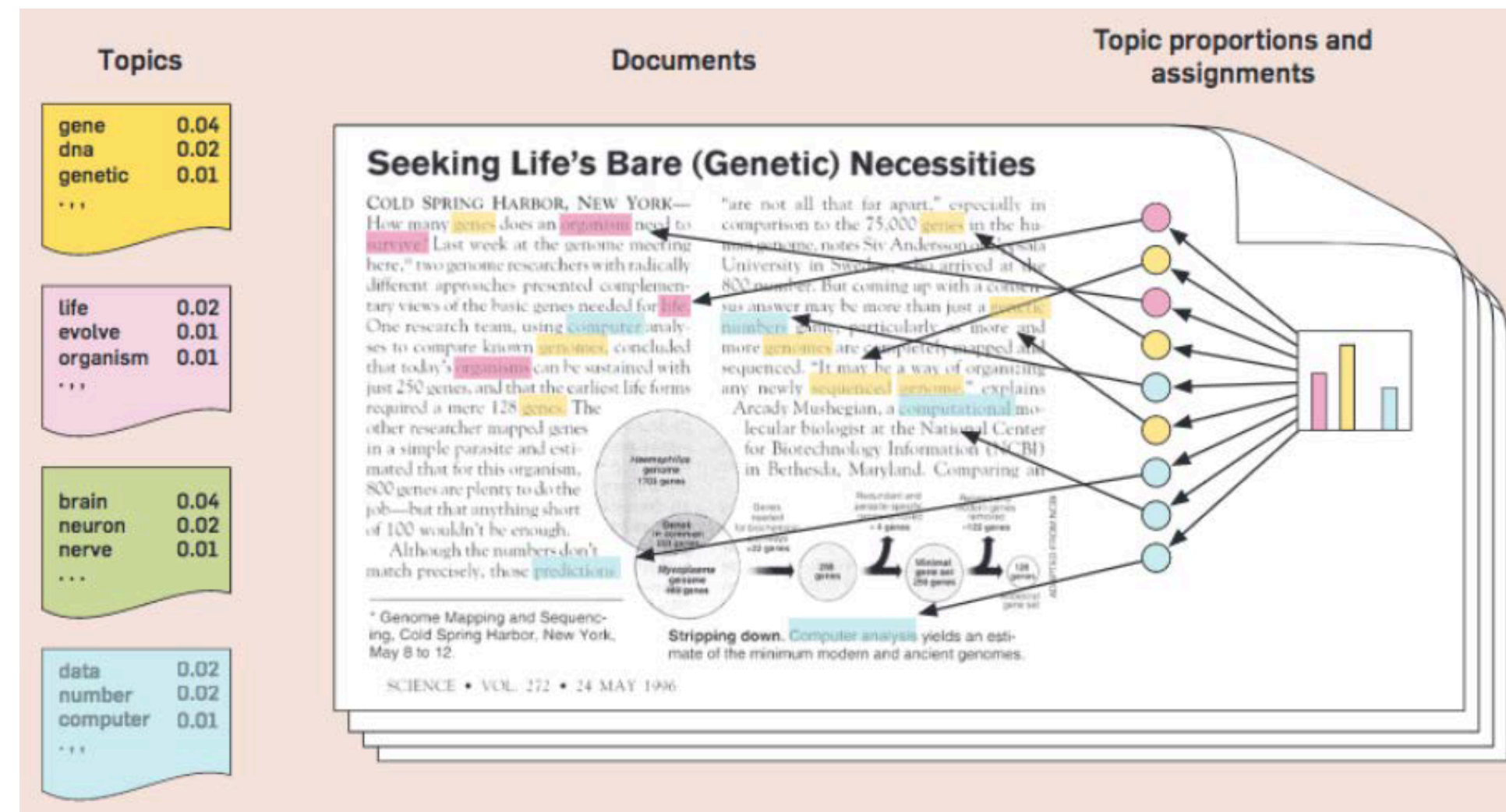
DATA 641

Lecture 13 — Necessary math for topic modeling

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Topic modeling

Topic modeling, a fundamental technique in natural language processing (NLP), helps uncover underlying themes in large collections of text data



The Significance of linear algebra in Topic Modeling

Linear algebra plays a crucial role in various aspects of topic modeling

- Document-Term Matrix (DTM): Topic modeling often starts with creating a DTM. Each row represents a document, each column represents a term, and each cell contains term frequency. Linear algebra operations are used for DTM analysis.
- Matrix Factorization: Techniques like Singular Value Decomposition (SVD) and Non-Negative Matrix Factorization (NMF) are used to factorize the DTM. Linear algebra helps in extracting latent topics from the matrix.
- Eigenvalues and Eigenvectors: Eigenvalues and eigenvectors provide insights into the structure of the textual data.
- Dimensionality Reduction: Principal Component Analysis (PCA) is an example used for this purpose.
- Probabilistic Models: In probabilistic topic models like Latent Dirichlet Allocation (LDA), linear algebra is employed for inference and parameter estimation.

Vectors

For a 1—dimensional array of real numbers, with dimension n , $\boldsymbol{x} \in \mathbb{R}^n$

Some common vector operations:

- Scalar product:

$$\alpha \boldsymbol{x} = [\alpha x_1, \alpha x_2, \dots, \alpha x_n] \quad (1)$$

- Dot (inner) product:

$$\boldsymbol{x} \cdot \boldsymbol{y} = \boldsymbol{x}^\top \boldsymbol{y} = \sum_{i=1}^n x_i y_i \quad (2)$$

- Norm:

$$\|\boldsymbol{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \quad (3)$$

Matrices

For a 2—dimensional array of real numbers, with dimension $m \times n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$

Some common vector operations:

- Pointwise operations: Multiplication, scalar multiplication, addition, subtraction
- Matrix-Vector Multiplication (makes a vector):

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix} \quad (4)$$

Matrices

- Matrix-Matrix Multiplication $\mathbf{AB}$ (makes a matrix):

$$\begin{aligned}\mathbf{AB} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} & a_{11}b_{13} + a_{12}b_{23} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} & a_{21}b_{13} + a_{22}b_{23} \\ a_{31}b_{11} + a_{32}b_{21} & a_{31}b_{12} + a_{32}b_{22} & a_{31}b_{13} + a_{32}b_{23} \end{bmatrix}\end{aligned}\tag{5}$$

- Matrix-Matrix Multiplication $\mathbf{BA}$ (makes a matrix) not necessarily same as the previous one:

$$\begin{aligned}\mathbf{BA} &= \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \\ &= \begin{bmatrix} b_{11}a_{11} + b_{12}a_{21} + b_{13}a_{31} & b_{11}a_{12} + b_{12}a_{22} + b_{13}a_{32} \end{bmatrix}\end{aligned}\tag{6}$$

Matrices

For a 2—dimensional array of real numbers, with dimension $m \times n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, we define the following

- Identity $n \times n$ matrix

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \quad (7)$$

- Inverse matrix

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (8)$$

- The transpose of a matrix $\mathbf{A}$, denoted as $\mathbf{A}^\top$, is obtained by swapping its rows and columns.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^\top = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \quad (9)$$

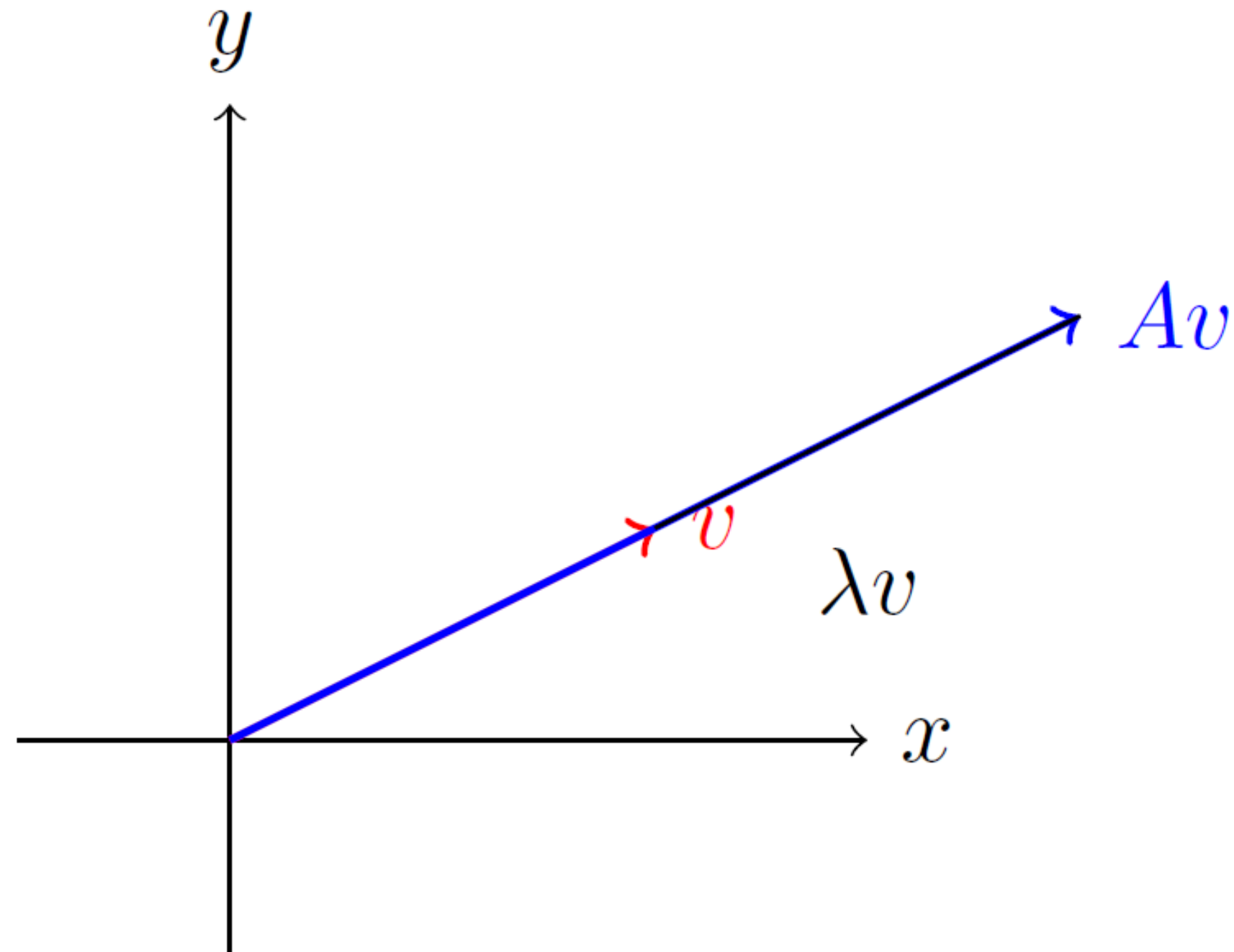
Eigenvalues and Eigenvectors of a Matrix

- Eigenvalues and Eigenvectors are fundamental concepts in linear algebra and have important applications in NLP and especially in topic modeling.
- In the context of a square matrix $\mathbf{A}$ of size $n \times n$:
 - **Eigenvalue (λ):** An eigenvalue of $\mathbf{A}$ is a scalar that represents how the matrix scales or stretches space in a particular direction.
 - **Eigenvector ($\mathbf{v}$):** An eigenvector of $\mathbf{A}$ is a non-zero vector that remains in the same direction after being scaled by $\mathbf{A}$. It characterizes the direction of stretching or compression associated with a specific eigenvalue.
- Mathematically, for an eigenvalue λ and its corresponding eigenvector $\mathbf{v}$, the relationship is expressed as:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \tag{10}$$

Eigenvalues and Eigenvectors of a Matrix

- Geometrically, this means that when $\mathbf{A}$ is applied to $\mathbf{v}$, the result is a vector that points in the same direction as $\mathbf{v}$ but is scaled by the factor λ .



Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique commonly used in data analysis and machine learning. It helps identify the most important features or components of a dataset while reducing its dimensionality.

Objective of PCA

PCA aims to transform a dataset with multiple correlated features into a new coordinate system, where the features (variables) are uncorrelated or orthogonal. This new coordinate system is defined by a set of axes called principal components. The primary goal is to capture as much of the original data's variance as possible in a reduced number of components.

Linear Algebra in PCA

1. Mean Centering:

- The first step in PCA is to subtract the mean (average) of each feature from the data points. This ensures that the new coordinate system is centered at the origin.
- In matrix form, if we have a dataset $\mathbf{X}$ with n data points and p features, we compute the mean-centered data matrix as follows: $\mathbf{X}_{\text{centered}} = \mathbf{X} - \bar{\mathbf{X}}$ Here, $\bar{\mathbf{X}}$ is the mean of each feature.

2. Covariance Matrix:

- PCA involves computing the covariance matrix of the mean-centered data. The covariance matrix summarizes the relationships between different features.
- The covariance matrix $\mathbf{C}$ of $\mathbf{X}_{\text{centered}}$ is calculated as follows:
$$\mathbf{C} = \frac{1}{n-1} \cdot (\mathbf{X}_{\text{centered}})^\top \cdot \mathbf{X}_{\text{centered}}$$
 This matrix describes how each feature varies with respect to the others.

3. Eigenvalue Decomposition:

- The next step is to find the eigenvalues and eigenvectors of the covariance matrix $\mathbf{C}$. This involves solving the linear algebraic equation: $\mathbf{C} \cdot \mathbf{v} = \lambda \cdot \mathbf{v}$
 - $\mathbf{v}$ represents the eigenvector, which defines the direction of a principal component.
 - λ represents the eigenvalue, which quantifies the amount of variance explained by the corresponding principal component.

4. Selecting Principal Components:

- Principal components are ranked by the magnitude of their eigenvalues. The higher the eigenvalue, the more variance the corresponding component explains. Typically, a subset of the top principal components is selected to reduce dimensionality while retaining most of the dataset's variance.

5. Projection:

- Finally, the dataset is projected onto the selected principal components to obtain the reduced-dimensional representation.
- The projection of the mean-centered data $\mathbf{X}_{\text{centered}}$ onto k principal components is calculated as: $\mathbf{X}_{\text{reduced}} = \mathbf{X}_{\text{centered}} \cdot \mathbf{V}_k$ Where $\mathbf{V}_k$ contains the top k eigenvectors as columns.

PCA's geometric interpretation involves rotating and projecting data points to reduce dimensionality while preserving essential patterns and variations. It simplifies complex datasets and aids in visualizing data relationships and structures.

Topic Identification using PCA

- In the reduced-dimensional space, each document is represented as a vector in the k -dimensional subspace defined by the selected principal components. These vectors represent the documents in a new basis that captures the most significant variations in the data.
- We can apply clustering algorithms (e.g., K-Means or DBSCAN) or probabilistic topic modeling techniques (e.g., Latent Dirichlet Allocation or Non-Negative Matrix Factorization) to identify topics based on the reduced-dimensional representations.
- Topics can be characterized by the words and documents that are most strongly associated with each principal component.